Verification record

Verified on **2026-10-16 UTC** on the shared Windows 11 development machine. These are measured results, not promises for every machine or dataset.

Delivered runtime

- Portable CPython **3.13.16**, SQLite **3.50.4**.
- Bundled Prometheus **3.15.0**, with four validated alert rules.
- Official runtime SHA-256 checks verified before extraction.
- A clean rebuild using all hash-pinned wheels in `requirements.lock` succeeded.
- Unused pip-generated executables containing build-host interpreter paths were removed. Native tools and Python `-m` entry points remain functional.
- Loaded Visual C++ runtime DLLs used by Python/Arrow were located in the bundle; the imported process did not require an installed `msvcp140` runtime.

Real data

The actual EEA import and conditional refresh succeeded without registration.

Item	Verified value
Measurement files processed per complete cycle	87
Stored source observations after refresh	216,769
Derived physical-station/day records	12,187
EEA city candidates	17
Physical stations retained	49
Stations without an official city mapping, excluded	12
Queryable completed history	2025-10-16 through 2026-10-15, inclusive UTC

Source-validity, missing-value and negative-value exclusions were counted and logged, not converted into zero pollution. Repeated imports did not duplicate stored observations. Actual full-year results used three eligible PM2.5 cities and thirteen eligible PM10 cities under the default coverage policy.

These counts describe this snapshot. Source updates can change counts, coverage and results. They are not a claim of complete national monitoring.

Automated tests

Ruff passed; 60 offline tests passed, including in the relocated project via the packaged `Test-CleanAir.cmd`.

Coverage includes:

- station/day weighting and distinct physical station identity;

- exact daily-reference boundaries, custom thresholds and unrounded ranking;
- pollutant selection, ties and insufficient coverage;
- 75% hourly and 90% period eligibility boundaries;
- invalid/reversed/future dates and nonfinite or missing custom thresholds;
- SQLite persistence, WAL, indexes, revisions and source precedence;
- actual Parquet schemas, units, aggregation windows and missing identifiers;
- HTTP retries, conditional downloads and file-level replay;
- a real collector failure path followed by successful cached API queries;
- no third-party requests during comparisons;
- exact 36-month archive boundary, verified round-trip restoration, corruption rejection and no active-data pruning after archive verification failure;
- consistent database backup and handover ZIP contents;
- runtime checksum verification and safe extraction;
- explicit API errors, offline HTML/API documentation, output escaping and host validation;
- correct offline status even when the previous stored collection message described online updates.

One upstream **Starlette TestClient deprecation warning** is emitted for its current HTTPX integration. It does not fail tests or affect production requests. The dependency lock prevents an unreviewed framework upgrade; future upgrades should address that test-client transition and rerun the suite.

Complete HTTP performance

Target: **p95 below 300 ms**. Each case ran **100 successful, shape-validated requests with four concurrent clients** against the actual HTTP service and real stored data.

Machine: Windows 11 x64, Intel64 Family 6 Model 170, 22 logical CPUs, Python 3.13. Other activity on the shared machine was not controlled.

Pollutant	Period	HTTP p95
PM2.5	One day	66.571 ms
PM2.5	Thirty days	60.741 ms
PM2.5	Twelve months	60.715 ms
PM10	One day	78.140 ms
PM10	Thirty days	63.425 ms
PM10	Twelve months	136.557 ms

All six cases met the target. No query-result cache was introduced to obtain these results. Daily rollups and indexes were sufficient.

Full evidence:

- Complete HTTP measurements
- Isolated comparison-service measurements

The latter benchmark is deliberately separate: in-process timings are not misrepresented as complete HTTP timings.

Browser checks

- Loaded the actual service and checked desktop/mobile layouts at 1440, 390 and 260 CSS pixels, with no document-level horizontal overflow.
- Submitted a full-year PM10 comparison through the form.
- Changed WHO guidance to a custom daily threshold of zero. The displayed winner and mean stayed **Vratsa / 11.88 ug/m3**, while days within the reference changed from **99.7% to 0.3%**.
- Confirmed reversed dates receive HTTP 422 and an accessible, useful error panel.
- Confirmed all automatically loaded application resources are same-origin.
- Confirmed offline status remains accurate after client-side status polling.

The example winner is only for that pollutant, date range, snapshot and policy. It is not an unconditional statement about where to live.

Monitoring and portability checks

1. Started the complete original application and monitoring; both returned ready.
2. Prometheus successfully scraped `/metrics`; its generated configuration and all four alert rules passed `promtool`.
3. Attempted handover during an active application process. It was rejected with an explicit workspace-lock error; the running application remained alive.
4. Generated a ZIP using `Prepare-Handover.cmd`; the packager verified its SQLite snapshot and every ZIP entry's CRC.
5. Extracted and renamed the project to a different folder containing spaces.
6. Ran the copied interpreter's `doctor` from a different working directory: it resolved the new workspace, reported database integrity `ok`, and retained the complete historical range.
7. Ran the **actual copied** `Start-CleanAir.cmd` with `--offline --no-browser --port 8001`, without using installed Python.
8. Confirmed offline API results remained correct and Prometheus automatically scraped `127.0.0.1:8001`, not the original port.
9. Ran all sixty tests through the copied Windows test launcher. The final offline-status correction was also checked in that relocated project.
10. Used the relocated UI on a mobile viewport and confirmed configurable comparisons worked with no external application resources.

The relocation check used the same Windows machine, not a newly provisioned VM. Corporate execution policy and other hardware/OS combinations were not tested. The final ZIP includes a SHA-256 sidecar; regenerate and recheck it after changes.

Stop launcher follow-up

Added `Stop-CleanAir.cmd` and a workspace-local cooperative shutdown channel. The stop-launcher update passed **68 tests**, with Ruff passing. The existing upstream TestClient deprecation warning remains non-blocking.

An actual Windows start/stop exercise verified:

- a running application and Prometheus shut down cleanly, without forced process termination;
- running the stop launcher again succeeds when the application is already stopped;

- an orphaned interpreter from the workspace is terminated and its resources released;
- an unrelated interpreter outside the workspace remains running;
- SQLite is checkpointed, database integrity is checked, and critical runtime, database, log and monitoring files can be opened without sharing conflicts.

New regression tests also cover stale stop requests, shutdown metadata cleanup, active workspace-lock refusal, existing Ctrl+C behaviour and explicit failed-start errors. The handover-content test requires both the stop launcher and its helper.

Complete Git LFS checkout follow-up

The runtime files and measurement database are now part of `main`, with large binaries and SQLite stored through Git LFS. Runtime files retain their original bytes without Git newline conversion. Python extension modules (`.pyd`) are included; only disposable bytecode caches are excluded.

The current complete suite contains **69 passing tests**, with Ruff passing. The additional test verifies that the Windows start launcher explains an unresolved LFS pointer instead of trying to execute it as Python. The database was checkpointed and integrity-checked before preparing the tracked snapshot.

Git LFS must retrieve its objects for a runnable Git checkout. The previously published portable Release remains available as a complete ordinary ZIP.